

Master’s Thesis Proposal

Harvard Master’s in Data Science

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1. Title

Attention-Guided Router Steering: A Feature-Learning Lens on Behavioral Intervention in Sparse Mixture-of-Experts Language Models.

2. Advisor

Adityanarayanan (Adit) Radhakrishnan — Assistant Professor, MIT Department of Mathematics; Associate Member, Broad Institute of MIT and Harvard (aditradha.com). A Harvard SEAS co-advisor will be added if the program requires an internal sponsor when the primary advisor is external; pending confirmation with Daniel Weinstock.

3. Abstract

Sparse mixture-of-experts (MoE) language models such as Mixtral 8x7B route every token to a small subset of feed-forward experts via a learned router. Recent work (STEERMOE) showed that biasing the router logits of selected experts can steer generations toward a target concept at inference time. In a faithful re-implementation I find that this recipe transfers *weakly* to Mixtral on a question-only concept-transfer test, and that the underlying signal is diffuse: the top twenty (layer, expert) cells out of $32 \times 8 = 256$ capture less than 10% of the total $|\Delta|$ mass. The thesis proposes **Attention-Guided Router Steering (AGRS)**, a drop-in replacement for STEERMOE in which the per-token contribution to the readout is weighted by attention mass to a concept-bearing prefix span. AGRS reduces to STEERMOE when the attention weights are uniform, isolating the contribution to the readout step. I evaluate AGRS against STEERMOE and three hidden-state steering baselines across four MoE checkpoints and four concept families, with a pre-registered decision gate before the more expensive runs, and use the Average Gradient Outer Product (AGOP) framework from my advisor’s recent work to ask whether AGRS’s expert selections are feature-aligned in a way STEERMOE’s are not.

4. Motivation and Background

My prior research is a Harvard project on hyperbolic prototype embeddings for hierarchical artistic style on WikiArt,¹ which trained the same prototype classifier in matched-dimension Euclidean and Poincaré-ball output spaces across 150 seed-replicated runs. That project taught me how to design a tightly scoped representation study, run an honest mixed-results comparison, and ship a reproducible end-to-end artifact — skills directly transferable here.

For the past several months I have been running STEERMOE-style routing-bias experiments on Mixtral 8x7B-Instruct on the MIT ORCD cluster.² The preliminary finding that motivates the thesis is empirically concrete. Across 25 question-only test cases on Mixtral, swept across eight (α, k_+, k_-) STEERMOE configurations, the concept-token hit rate moves by at most 0.04 between baseline and STEERMOE; response length contracts monotonically with the steering coefficient (-2.16 to -5.52 words); and the per-(layer, expert) risk-difference signal is diffuse, with the top twenty cells out of 256 capturing less than 10% of the total $|\Delta|$ mass. If the readout is diffuse and the intervention barely moves the model, then reweighting the readout by attention to the concept span is the lightest-touch method change that could plausibly fix it.

This is also where the project meets my advisor’s research. Adit’s recent work on the Average Gradient Outer Product (AGOP) — *Mechanism for Feature Learning in Neural Networks* (Science, 2024) and follow-ups — introduces AGOP as a mathematical handle on *which input directions a model treats as informative for a given prediction*. For sparse MoE routing the analogous question is *which input directions a router’s gating function for expert e at layer ℓ treats as informative*, and the thesis tests whether AGRS picks experts whose top AGOP eigenvector aligns with concept-prototype directions in a way STEERMOE does not.

After this master’s I intend to pursue research roles in AI interpretability and behavioral evaluation, with industry research labs working on alignment and post-training intervention methods as the primary target.

5. Goals and Expected Outcomes

1. **Method.** Define and implement AGRS as a drop-in replacement for STEERMOE on Mixtral 8x7B and OLMoE, with unit tests including the property test that AGRS reduces to STEERMOE when attention weights are uniform. *Success criterion:* AGRS produces strictly more-concentrated risk-difference tables than STEERMOE on $\geq 4/5$ pilot concepts.
2. **Empirical evaluation.** A pre-registered comparison of AGRS against five alternatives (baseline, STEERMOE, ITI, ActAdd, RepE difference-of-means) across four MoE checkpoints and four concept families, $n = 250$ cases per condition, five seeds, with concept-transfer rate from an LLM judge cross-checked against a 200-case human-rater subset as the primary metric. *Success criterion:* AGRS lifts CTR by ≥ 0.10 over STEERMOE on Mixtral / fears with non-overlapping paired bootstrap 95% CIs, and produces a positive shift in $\geq 3/4$ models and $\geq 3/4$ concept families.
3. **Mechanistic analysis.** Estimate the AGOP of selected experts’ gating functions and ask whether AGRS’s selections align with concept-distinctive directions. *Success criterion:* AGRS-selected

¹github.com/pgrindehollevik-harvard/hyperbolic.

²Working draft and code: github.com/pgrindehollevik-harvard/MoE_attention_guided_steering (paper-submission branch).

experts have systematically higher cosine alignment between their top AGOP eigenvector and the concept-prototype embedding than STEERMoE-selected experts; the per-cell alignment-vs-CTR correlation across (model, concept) cells reaches $|r| \geq 0.5$.

Deliverables. A submitted NeurIPS or ICLR main-track paper, a thesis monograph organized around the three goals above, and an anonymized public artifact tree under CC-BY (executed notebooks, generation metadata, qualitative reviews, activation tables, steering plans, judge transcripts, Apptainer image, pre-registration document).

6. Scientific Justification

Behavioral steering at inference time is one of the few corners of mechanistic interpretability where small interventions on frozen models produce measurable, causal behavioral effects. Most published steering methods operate on the residual stream of dense transformers; sparse MoE models give us a structurally different intervention site — the router logit — whose effect is mediated through a discrete selection over a small expert pool. Whether the readout-and-intervene paradigm transfers from dense to sparse architectures, and whether it does so with the same readout rules, is open. The open-weight frontier is increasingly MoE (Mixtral, OLMoE, Qwen2-MoE, DeepSeek-MoE), so the answer has direct deployment relevance for alignment researchers working on those checkpoints.

The contribution is innovative in three specific ways. (i) *The method.* AGRS is, to my knowledge, the first attention-guided readout developed for router-logit interventions on sparse MoEs; the reduction to STEERMoE under uniform attention isolates the readout step from confounding implementation choices. (ii) *The mechanistic framing.* AGOP has not been applied to MoE routing-bias steering; doing so either validates the feature-learning lens on routing decisions or constrains where the framework applies. (iii) *The empirical rigor.* A pre-registered Phase-3 decision gate ($\Delta\text{CTR} \geq +0.10$) is set before any AGRS-vs-STEERMoE data is collected, so the headline cannot be backfit to the result.

7. Literature Review

Activation steering on dense transformers — Inference-Time Intervention (Li et al., NeurIPS 2023), Representation Engineering (Zou et al., 2023), Activation Addition (Turner et al., 2023), and Attention-Guided Steering (Tan et al., 2024) — has established that small additive interventions on the residual stream can causally shift behavior; the last of these is the direct upstream inspiration for the readout half of AGRS, none of these methods target sparse MoE routing. Sparse MoE language models — Mixtral (Jiang et al., 2024), OLMoE (Muennighoff et al., 2024), Qwen2-MoE, DeepSeek-MoE — provide the open-weight substrate used here. STEERMoE (2024) is the upstream router-bias method this thesis builds on; its uniform-readout rule is what AGRS replaces. Finally, the feature-learning line developed by my advisor — the Average Gradient Outer Product (*Science*, 2024), its CNN extension (arXiv:2309.00570), and Recursive Feature Machines (arXiv:2401.04553) — supplies the mathematical machinery I use to test whether AGRS’s expert selections are feature-aligned. The thesis sits at the intersection of the third and fourth threads.

8. Methodology

The project is mainly computational. Every claim is empirical and reproducible from a pinned codebase, a pinned environment, and a fixed set of model checkpoints.

The AGRS readout. For a contrastive prompt pair (p^+, p^-) with concept-bearing prefix span $\mathcal{C} \subset p^+$, let $R_{\ell,t,e} \in \{0,1\}$ indicate that expert e is selected by Mixtral top-2 routing at layer ℓ , position t ; let $a_t = \max_h \sum_{t' \in \mathcal{C}} A_{h,t \rightarrow t'}^{\ell^*}$ be the per-token attention mass to the concept span at a chosen reference layer ℓ^* . The AGRS attention-weighted activation rate is

$$r_{\ell,e}^+ = \frac{\sum_t a_t^+ \cdot R_{\ell,t,e}^+}{\sum_t a_t^+}, \quad \Delta_{\ell,e} = r_{\ell,e}^+ - r_{\ell,e}^-$$

with $a^- \equiv 1$ on the unprefixed prompt. The decode-time intervention is identical to STEERMOE: add $\pm\alpha$ to the top- k_+ positive and top- k_- negative experts’ router logits. STEERMOE is recovered exactly when $a_t^+ \equiv 1$.

Pipeline. Per-token router logits and attention weights are captured via Hugging Face forward hooks, sliced to the shared statement-body target span, aggregated into the risk-difference table, used to build a sparse plan, and re-injected at decode time. Generations are scored by an LLM judge (GPT-4o as primary, Claude as cross-check) on a 0–2 ordinal scale for thematic concept transfer, with a human-rater subset of 200 cases for judge calibration. Statistical reporting is paired bootstrap 95% CIs across seeds and paired t -tests on seed-collapsed deltas.

Mechanistic analysis. For each selected expert e at layer ℓ , estimate $M_{\ell,e} = \mathbb{E}_x [\nabla_x g_{\ell,e}(x) \nabla_x g_{\ell,e}(x)^\top]$ of the gating function over the contrastive prompt pool, and report the cosine between its top eigenvector and the concept-prototype embedding direction. The empirical question is whether AGRS picks better-aligned experts than STEERMOE.

Reproducibility. Every experimental step is a Jupyter notebook with a `papermill` parameters cell, run identically on a laptop in smoke mode and on the MIT ORCD cluster via a paired `sbatch` wrapper. All generation calls are at temperature 0 with explicit seeds; every model load asserts a recorded Hugging Face revision SHA; every run writes a `provenance.json` with the git SHA, cluster job id, and wallclock. The container is a single Apptainer image; the AGRS hypothesis is pre-registered on OSF with the decision gate before the gate-determining experiment runs.

Models and compute. Mixtral 8x7B-Instruct (8-bit with CPU offload), OLMoE 1B-7B-Instruct, Qwen2-MoE 14B, and DeepSeek-MoE-Lite if licensing allows; total budget ~ 305 GPU-hours on the MIT ORCD `mit_normal_gpu` partition over the academic year.

Timeline.

Months	Milestone
May–Jun 2026	Reproducibility hardening; <code>make paper</code> on clean checkout.
Jun–Jul 2026	Evaluation harness; judge–human Spearman ≥ 0.7 .
Jul 2026	AGRS implementation; unit tests; pre-registration filed.
Aug 2026	Pilot run; Phase-3 decision gate evaluated.
Sep–Oct 2026	Full six-method comparison on Mixtral / fears.
Nov 2026	Cross-model generalization (Mixtral, OLMoE, Qwen2-MoE, DS-MoE).
Dec 2026	Cross-concept generalization (fears, professions, refusal, sentiment).
Jan 2027	AGOP mechanistic analysis.
Feb–Mar 2027	Statistics, thesis monograph, conference submission.

If the Phase-3 gate fails, the pre-registered pivot is either an attention-weighted hidden-state intervention (testing whether the readout is the bottleneck independently of the intervention site) or a focused diagnostic-and-AGOP study of why the routing signal is diffuse on Mixtral.